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Project Title:	Computer Modeling of the Role of Capillary Transport in Producing Postflight Orthostatic Intolerance		
Division Name:	Physical Sciences		
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Comments:			
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Postflight orthostatic intolerance (POI), which describes the dizziness astronauts feel upon standing after returning from space, is a common problem. POI compromises crew safety and could have disastrous consequences in emergency situations during landing. A potentially revealing characteristic of POI is that even susceptible astronauts are able to stand postflight for several minutes without symptoms. Many astronauts become dizzy only after 5-10 minutes of standing. Therefore, compensatory reflexes are apparently sufficient to prevent symptoms initially. While many factors, including a well-documented decrease in postflight blood volume, may increase sensitivity, the final cause of POI must have a delayed action consistent with the range of times astronauts are able to tolerate standing. Onset of symptoms during orthostatic stress in nonastronauts is correlated with the gradual rate of loss of blood volume by capillary filtration, which provides a potential explanation for the delayed occurrence of POI. Furthermore, astronauts have increased lower body capillary permeability in space. If permeability remains elevated, then capillary transport may also

Task Description:	be responsible for the reduced tolerance of astronauts to upright posture postflight. Accordingly, it is hypothesized that capillary filtration explains the delayed onset, as well as the increased postflight incidence of dizziness. The hypothesis will be tested with a computer model of the cardiovascular system that incorporates fundamental biomechanical responses and reflex control. Posture-dependent capillary filtration will be represented with local permeability values and models of local intra- and extra-vascular pressure. A model of cerebral circulation will be included to provide a prediction of dizziness by direct comparison with established thresholds for cerebral flow. Because data is currently unavailable in astronauts, the model will be validated with heart rate, stroke volume and arterial pressure measurements and capillary filtration estimates derived from previous and soon to be conducted experiments on nonastronauts. Finally, the plausibility of the hypothesis will be tested by comparing the permeability required in the model to cause dizziness during simulated stand tests to the preliminary inflight values. The results of this project will help justify and guide the design of experiments to investigate this mechanism in astronauts, as well as to formulate and evaluate countermeasures. The overall objective of this research is to investigate the role of increased capillary transport in producing POI. The specific aim of the project is to simulate the influence of capillary transport on cerebral perfusion with a computational model of the cardiovascular system that incorporates important primary biomechanical responses and secondary neurohumoral responses. - Capillary permeability will be added to a current computer model of cardiovascular adaptations to gravity. Concurrently, several improvements will be made to the model to accurately simulate the influence of reduced volume on cerebral perfusion. Baseline capillary permeability values will be derived from LBNP tests of nonastronauts not subjected to simulated microgravity. These experiments will also provide time-dependent heart rate, arterial pressure and stroke volume data for validating the code. Stand tests of astronauts will then be simulated with increases in capillary permeability to levels high enough to reduce cerebral perfusion sufficiently to cause syncope. These increases will be compared to those found in astronauts, to determine whether capillary transport may be sufficiently increased to explain the prevalence of POI. POI is perhaps the most prevalent and potentially life-threatening cardiovascular problem related to spaceflight. This modeling effort will provide preliminary data necessary to justify focused experimental investigation of capillary transport. If these simulations indicate that this mechanism may be significant, then experiments to confirm the importance of capillary transport in astronauts and to investigate mechanisms of cellular adaptation will be pursued, and effective countermeasures can be designed and tested.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many, particularly elderly, patients on earth suffer from chronic orthostatic intolerance. Finding from this project may suggest new avenues of investigation that may contribute to more effective treatment of these patients.
Task Progress:	This project was initiated in April 2004. A graduate student was hired and work has begun to establish a set of baseline results for future modifications to the extensive cardiovascular model. In addition, development has begun on a simpler model that can provide faster, though rougher, estimates of cardiovascular response to a smaller set of parameter variations.
Bibliography Type:	Description: (Last Updated: 10/07/2004)
Abstracts for Journals and Proceedings	Sharp, M.K. "Computer modeling of cardiovascular responses to gravity" NASA Fluid Physics Conf, Cleveland, OH 1 , Jun-2004